

From Offline Unified Speech Modeling to Causal Incremental Generation: UniSS Phase 1-3, Streaming Stage A/B, and a Quality-Aware Reinforcement Learning Route

Jason Lee (Li Jin)

UniSS Training Reproduction and Simultaneous S2ST Project

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Abstract

This report presents a complete engineering and research route for extending an offline speech-to-speech translation (S2ST) system into a streaming or simultaneous S2ST system. The offline model uses Qwen2.5-0.5B-Instruct as a shared autoregressive backbone and represents text, GLM acoustic tokens, and BiCodec semantic tokens in a unified vocabulary. Training proceeds through Phase 1 foundational multitask alignment, Phase 2 direct S2ST with capability replay, and Phase 3 Quality/Performance specialization. The public UniST-198 reproduction contains 19,785,924 training records. The best Phase 3 checkpoint reaches Quality-mode Chinese-to-English/English-to-Chinese Text-BLEU scores of 39.38/48.17 and Speech-BLEU scores of 22.73/46.31 on the UniST test set. Relative to Phase 2, Phase 3 improves 21 of 24 principal higher-is-better test cells.

The streaming route preserves the successful Phase 3 architecture and introduces two causal adaptation stages. Stage A connects a chunk-causal WhisperVQ frontend and an acoustic bridge to the shared Qwen model to learn append-only streaming ASR. Its 381-update training run is numerically stable, but its free-running weighted WER/CER remains 27.14%, so the formal quality gate is not passed. Stage B freezes the Stage A causal acoustic frontend and updates only the shared Qwen at a low learning rate using five task families: streaming ASR, incremental MT, end-to-end semantic deltas, and two Phase 3 replay modes. Stage B completes 2,264 updates. For English-to-Chinese at 160/320 ms decision chunks, it reduces first translated audio from 4.22 s to 2.40/2.88 s, corresponding to improvements of 1.82/1.34 s. These early outputs are nevertheless incomplete, while Chinese-to-English degrades because an incorrect irreversible ASR prefix propagates through MT and speech synthesis. We therefore do not claim that Stage B is a bidirectional overall success.

Building on these findings, we formulate a future quality-aware Group Relative Policy Optimization (GRPO) stage. For each input, the policy samples multiple READ/WRITE-content-audio trajectories and receives a reward combining final translation quality, prefix support, First WRITE, average token delay, completeness, non-silence, and boundary continuity, with reference-policy KL regularization. Earlier action-only GRPO experiments show why this extension is needed: a group-size-8 policy improves Chinese-to-English Text-BLEU by 1.87 over matched continued SFT, but increases First WRITE by 41.85 ms. Reward-v2 experiments provide preliminary evidence that bilingual quality terms and adaptive KL can reduce ATD by 24.7–46.5 ms without necessarily sacrificing both directions. The principal contributions of this work are a reproducible unified-token offline curriculum, Phase-3-anchored causal acoustic adaptation, trajectory-routed joint incremental training in a single shared decoder, and an evaluation framework that explicitly separates verified gains, failed gates, proxy latency, and proposed reinforcement learning.

Keywords: speech-to-speech translation; simultaneous translation; streaming ASR; incremental machine translation; speech tokens; multitask learning; GRPO; quality-latency trade-off

1 Introduction

Direct speech-to-speech translation aims to map the linguistic content, language, and expressive characteristics of source speech into target-language speech. An offline system may observe the complete utterance before planning the translation and target waveform. A simultaneous system must instead make irreversible decisions before the future source context is available. Let the 16-kHz source waveform be the vector $s \in \mathbb{R}^N$, where N is the number of samples and $N/16000$ is the duration in seconds. An offline system estimates $p_\theta(\mathbf{y} \mid s)$, where $\mathbf{y} = (y_1, \dots, y_L) \in \mathcal{V}^L$ is a target token sequence of length L and $\mathcal{V}$ is the unified vocabulary. At arrival event k , a streaming system may only access the prefix $s_{1:n_k} \in \mathbb{R}^{n_k}$ and must emit an append-only increment $\Delta \mathbf{y}_k$. Previously emitted increments may not be revised. This creates three coupled conflicts: insufficient evidence versus premature output, waiting versus interaction latency, and fragmented generation versus speech continuity.

Our offline foundation follows the unified expressive S2ST formulation of UniSS[1]. It uses a Qwen2.5[7] Transformer to model text, source speech representations, and target BiCodec semantic tokens in a single autoregressive space. Training is implemented with Megatron-LM[5] and covers all 198 public UniST training shards. The streaming route draws on the READ/WRITE abstraction of prefix-to-prefix and wait- k translation[2] and on the multitask, multi-chunk motivation of StreamSpeech[8], while deliberately retaining the successful Phase 3 architecture and codec.

This work addresses four empirical questions:

1. Can a shared 0.5B autoregressive model reliably acquire ASR, translation, and speech-generation capabilities through a Phase 1-3 curriculum on the full public dataset?
2. Can a chunk-causal acoustic frontend support append-only streaming ASR without discarding the best Phase 3 checkpoint?
3. After freezing that frontend, can the shared Qwen jointly learn ASR deltas, incremental translation deltas, and target semantic deltas while retaining offline capabilities?
4. Can trajectory-level reinforcement learning improve the quality-latency trade-off that supervised trajectory learning does not optimize directly?

The main technical contributions are as follows.

- **A three-phase offline curriculum in a unified token space.** Text, GLM acoustic tokens, and BiCodec semantic tokens are modeled by the same 24-layer, 896-dimensional Qwen backbone. Phase 2 uses a 2:1 mixture of new tasks and Phase 1 replay, followed by Phase 3 Quality/Performance specialization. Phase 3 improves 21 of 24 principal test cells over Phase 2.
- **Phase-3-anchored causal acoustic transfer.** Stage A does not train a separate streaming system from scratch. It inserts a chunk-causal WhisperVQ frontend and bridge into the Phase 3 model and retains exact Phase 3 replay, making the origin of causal adaptation and capability retention auditable.
- **Five-family incremental training in one shared decoder.** Stage B combines streaming ASR, incremental MT, interleaved S2ST semantic deltas, and two Phase 3 replay modes in a globally synchronized schedule. Gold and free-running histories, alignment confidence, and clean/noisy/quarantine strata control supervision reliability.
- **Completeness-aware END margin and commit consistency.** A logit margin

discourages premature semantic END, while a consistency term protects already committed prefixes. The resulting English-to-Chinese first-audio improvement is measurable, but the failed Chinese-to-English gate reveals the cost of aggressive irreversible commits.

- **A quality-aware trajectory-level RL formulation.** Rather than optimizing only a binary WAIT/WRITE head, the proposed reward jointly measures final text and speech quality, prefix safety, latency, completeness, non-silence, and audio continuity. We state matched-SFT and Pareto-frontier criteria that must be met before an RL gain can be claimed.

2 Related Work

2.1 Unified Expressive Speech-to-Speech Translation

UniSS represents source acoustics, intermediate text, and target speech semantics in a unified language-model vocabulary, with Quality and Performance protocols providing different generation paths[1]. Our Offline Phase 1-3 is a public-data, 0.5B reproduction of this principle; it is not an exact reproduction of the original 1.5B model and its complete training corpus. Whisper provides a strong acoustic representation through large-scale weak supervision[3]. Qwen2.5 supplies the shared autoregressive language-model backbone[7]. BiCodec/Spark-TTS-style decoupling of global speaker tokens and local semantic tokens supports target speech generation in the source speaker’s voice[6].

2.2 Streaming Translation and Simultaneous S2ST

STACL formulates simultaneous translation as prefix-to-prefix generation and uses wait- k to control latency[2]. A fixed policy is simple but cannot adapt its waiting behavior to the ambiguity of a particular source prefix. StreamSpeech combines a streaming encoder, multitask learning, multi-chunk training, and CTC-based counting of source and target units[8]. We adopt its motivations that one model should cover multiple latency operating points and that ASR/MT auxiliary objectives should protect end-to-end S2ST. We do not directly reproduce its non-autoregressive unit CTC decoder: target speech remains a sequence of BiCodec semantic tokens generated by the Phase 3 autoregressive LM head.

2.3 Reinforcement Learning for Group-Relative Trajectory Optimization

GRPO estimates a normalized relative advantage from multiple responses to the same input, avoiding a separately trained value critic and constraining the learned policy with reference-policy KL[4]. This is relevant to simultaneous S2ST because automatically generated trajectories provide only one approximate READ/WRITE target, whereas the actual objective is a Pareto trade-off among final translation, speech validity, completeness, and latency. However, our earlier action-only results demonstrate that RL is not automatically beneficial: an incomplete reward can select conservative trajectories that reduce premature WRITE by waiting longer.

3 Problem Formulation and System Overview

3.1 Unified Autoregressive Sequence Modeling

The logical vocabulary has size $V = 180,407$ and is padded to $V' = 180,480$ for Megatron. A training example concatenates task prompts, source representations, transcription, translation, and target semantic tokens into $\mathbf{z} = (z_1, \dots, z_T) \in \mathcal{V}^T$, where $T \leq 18,000$ is the packed sequence length. The embedding matrix $\mathbf{W}_{\text{emb}} \in \mathbb{R}^{V' \times d}$ maps each token to a vector of dimension $d = 896$. The shared Transformer f_θ produces $\mathbf{H} = f_\theta(\mathbf{z}_{<T}) \in \mathbb{R}^{T \times d}$. Its hidden vector at position t is $\mathbf{h}_t \in \mathbb{R}^d$, and the language-model matrix $\mathbf{W}_{\text{lm}} \in \mathbb{R}^{V' \times d}$ defines

$$p_\theta(z_t | \mathbf{z}_{<t}) = \text{softmax}(\mathbf{W}_{\text{lm}} \mathbf{h}_t). \quad (1)$$

For a binary supervision mask $\mathbf{m} = (m_1, \dots, m_T) \in \{0, 1\}^T$, the unified autoregressive cross-entropy is

$$\mathcal{L}_{\text{CE}}(\theta; \mathbf{z}, \mathbf{m}) = -\frac{1}{\sum_{t=1}^T m_t} \sum_{t=1}^T m_t \log p_\theta(z_t | \mathbf{z}_{<t}). \quad (2)$$

The individual tasks do not require independent neural heads. They are distinguished by task, mode, and language tokens; by the input segments; and by the locations selected by $\mathbf{m}$.

3.2 Offline and Streaming Conditions

The offline model reads a complete source sequence $\mathbf{x}_{1:S} \in \mathcal{V}^S$, where S is the number of source tokens. At streaming event k , only $\mathbf{x}_{1:s_k}$ is available, with $s_k \leq S$. The model emits $\Delta \mathbf{y}_k \in \mathcal{V}^{\ell_k}$, where ℓ_k is the number of target tokens newly committed at event k . The committed prefix is

$$\mathbf{y}_k^{\text{commit}} = \Delta \mathbf{y}_1 \oplus \Delta \mathbf{y}_2 \oplus \dots \oplus \Delta \mathbf{y}_k, \quad (3)$$

where $\oplus$ denotes sequence concatenation. Strict causal visibility requires $\Delta \mathbf{y}_k$ to depend only on $\mathbf{s}_{1:n_k}$ and previous commits, never on $\mathbf{s}_{n_k+1:N}$.

3.3 End-to-End Architecture

Figure 1 shows the inheritance from Offline Phase 1-3 to Streaming Stage A/B and the proposed RL stage. Gray blocks are frozen, green blocks are updated in the indicated supervised stage, and orange dashed blocks remain future work.

4 Methodology

4.1 Offline Phase 1: Foundational Multitask Alignment

Phase 1 first establishes the elementary mappings required by unified S2ST: source speech to transcription, source speech to translation, text plus speaker condition to target semantics, and transcription to translation as an MT proxy. Let the corresponding scalar losses be $\mathcal{L}_{\text{asr}}$, $\mathcal{L}_{\text{s2tt}}$, $\mathcal{L}_{\text{tts}}$, and $\mathcal{L}_{\text{mt}}$, each instantiated from equation (2) with a different mask. If $\boldsymbol{\pi}^{(1)} \in [0, 1]^4$ is the four-dimensional task-probability vector with $\sum_{j=1}^4 \pi_j^{(1)} = 1$, then

$$\mathcal{L}_{\text{P1}} = \mathbb{E}_{q \sim \text{Cat}(\boldsymbol{\pi}^{(1)})} [\mathcal{L}_q]. \quad (4)$$

The initial high-learning-rate run became unstable late in training. We therefore retained its 3,300-update checkpoint and performed 15,465 recovery updates with a lower learning rate, a fresh optimizer, and cyclic shuffle.

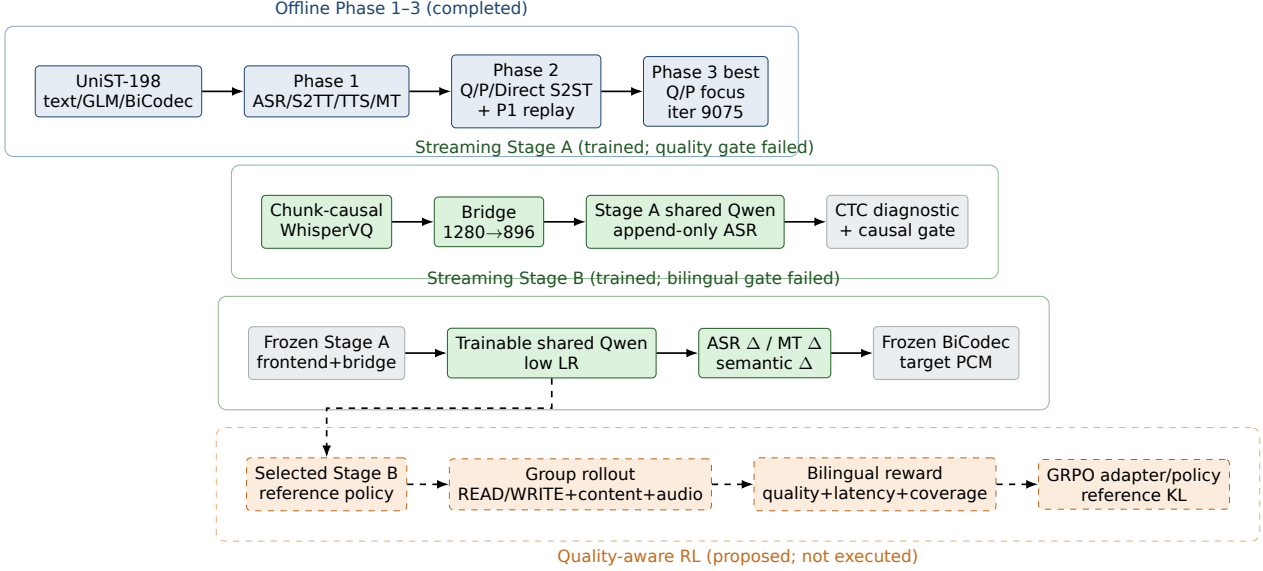


Figure 1: The complete UniSS offline-streaming-RL system. Stage A is initialized from the best Phase 3 checkpoint. Stage B freezes the causal acoustic frontend and updates the shared Qwen. Quality-aware RL is proposed only after the supervised quality gates are repaired.

4.2 Offline Phase 2: Direct S2ST with Capability Replay

Phase 2 introduces Quality, Performance, and Direct S2ST examples. Its purpose is to learn the complete text-to-speech output path without overwriting the ASR, S2TT, TTS, and MT capabilities obtained in Phase 1. Let $\mathcal{D}_{P2}$ and $\mathcal{D}_{P1}$ denote the Phase 2 and replay distributions. The 2:1 mixture is

$$\mathcal{L}_{P2} = \frac{2}{3} \mathbb{E}_{u \sim \mathcal{D}_{P2}} \mathcal{L}_{CE}(u) + \frac{1}{3} \mathbb{E}_{v \sim \mathcal{D}_{P1}} \mathcal{L}_{CE}(v). \quad (5)$$

Phase 2 loads only the Phase 1 model weights, not its optimizer or random-number-generator state, which makes this handoff an independently auditable stage.

4.3 Offline Phase 3: Quality/Performance Specialization

Phase 3 starts from Phase 2 checkpoint ‘iter_0015381’ and retains only the final Quality and Performance tasks. Let $\mathcal{D}_Q$ and $\mathcal{D}_P$ denote the two distributions and let $\rho \in [0, 1]$ be the Quality sampling probability. Then

$$\mathcal{L}_{P3} = \rho \mathbb{E}_{u \sim \mathcal{D}_Q} \mathcal{L}_{CE}(u) + (1 - \rho) \mathbb{E}_{v \sim \mathcal{D}_P} \mathcal{L}_{CE}(v). \quad (6)$$

The Quality path generally emits an explicit source transcription, target translation, and target semantic sequence. The Performance path is shorter and balances output quality with generation length. Phase 3 changes the task distribution rather than adding a new network module.

4.4 Streaming Stage A: Causal Acoustic Frontend and Incremental ASR

Phase 3 consumes a complete source GLM sequence and cannot directly accept incrementally arriving PCM. Stage A converts a waveform prefix into an 80-dimensional log-Mel matrix $\mathbf{X}_{1:t} \in \mathbb{R}^{t \times 80}$, where t is the number of currently visible acoustic frames. The chunk-causal WhisperVQ encoder produces $\mathbf{A}_{1:u} = g_\phi(\mathbf{X}_{1:t}, \mathbf{C}_{k-1}) \in \mathbb{R}^{u \times 1280}$, where

u is the visible encoded length and C_{k-1} is a cache containing only states from earlier chunks. The bridge maps the acoustic representation into the Qwen hidden dimension:

$$E^{\text{ac}} = \text{LN}(\mathbf{A})\mathbf{W}_b + \mathbf{b}_b, \quad \mathbf{W}_b \in \mathbb{R}^{1280 \times 896}, \quad \mathbf{b}_b \in \mathbb{R}^{896}, \quad (7)$$

yielding $E^{\text{ac}} \in \mathbb{R}^{u \times 896}$ for injection into Qwen.

The declared Stage A objective is

$$\begin{aligned} \mathcal{L}_A = & 1.00\mathcal{L}_{\text{AR-ASR}} + 0.30\mathcal{L}_{\text{CTC}} + 0.20\mathcal{L}_{\text{teacher-KL}} + 0.10\mathcal{L}_{\text{multi-chunk}} \\ & + 0.10\mathcal{L}_{\text{cache/full}} + 1.00\mathcal{L}_{\text{offline-ASR-replay}} + 1.00\mathcal{L}_{\text{P3-replay}}. \end{aligned} \quad (8)$$

$\mathcal{L}_{\text{AR-ASR}}$ is the append-only ASR-delta CE. For a source label sequence $\mathbf{a} \in \mathcal{V}^J$ of length J and a CTC posterior matrix $\mathbf{P} \in [0, 1]^{u \times (V_{\text{asr}}+1)}$, where V_{asr} is the ASR label-vocabulary size, CTC marginalizes all u -step blank-augmented paths:

$$\mathcal{L}_{\text{CTC}} = -\log \sum_{\pi \in \mathcal{B}^{-1}(\mathbf{a})} \prod_{r=1}^u P_{r, \pi_r}. \quad (9)$$

Here π is a length- u alignment path and $\mathcal{B}$ removes blanks and repeated labels. The multi-chunk term encourages compatible representations for the same audio at different chunks, while the replay terms preserve offline capabilities. Importantly, the formal run had a zero effective denominator for teacher KL, and cache/full consistency was implemented as an external gate. Neither term can therefore be credited with an observed gain.

The decision-chunk curriculum is defined by coverage progress $r \in [0, 1]$:

$$\mathcal{C}(r) = \begin{cases} \{1280, 960\} \text{ ms}, & 0 \leq r < 0.10, \\ \{960, 640\} \text{ ms}, & 0.10 \leq r < 0.35, \\ \{640, 320\} \text{ ms}, & 0.35 \leq r < 0.70, \\ \{320, 160\} \text{ ms}, & 0.70 \leq r \leq 1. \end{cases} \quad (10)$$

4.5 Streaming Stage B: Five-Family Joint Incremental Training

Stage A builds a streaming ASR interface but does not ensure incremental translation or continuous target speech. Stage B therefore freezes WhisperVQ, the bridge, and the CTC head, and updates only the shared Qwen. The maximum learning rates are 2×10^{-6} for the Transformer body and 5×10^{-7} for the tied embedding/LM head.

Each utterance is converted into a trajectory $\tau = (e_1, \dots, e_K)$ of K events. Event k is

$$e_k = (t_k, \Delta \mathbf{a}_k, \Delta \mathbf{b}_k, \Delta \mathbf{u}_k, q_k), \quad (11)$$

where $t_k \in \mathbb{R}_{\geq 0}$ is the source time, $\Delta \mathbf{a}_k \in \mathcal{V}^{J_k}$ is the ASR increment, $\Delta \mathbf{b}_k \in \mathcal{V}^{M_k}$ is the target-text increment, $\Delta \mathbf{u}_k \in \mathcal{V}^{U_k}$ is the target-semantic increment, and $q_k \in [0, 1]$ is alignment confidence. J_k , M_k , and U_k are the three increment lengths. Gold histories provide stable supervision, while free-running histories expose the model to its own prefix errors. Records are routed into clean, noisy-content, and quarantine strata. Quarantined examples may participate in Phase 3 replay but not in the strict incremental tasks.

The steady task mixture is 25% streaming ASR, 20% incremental MT, 30% end-to-end S2ST, 15% Phase 3 Quality replay, and 10% Phase 3 Performance replay. The active objective is

$$\begin{aligned} \mathcal{L}_B = & 1.00\mathcal{L}_{\text{ASR-}\Delta} + 1.00\mathcal{L}_{\text{MT-}\Delta} + 1.00\mathcal{L}_{\text{semantic-}\Delta} + 0.50\mathcal{L}_{\text{replay}} \\ & + 0.30\mathcal{L}_{\text{V1-ASR-KL}} + 0.25\mathcal{L}_{\text{P3-KL}} + 0.20\mathcal{L}_{\text{commit}} + 0.10\mathcal{L}_{\text{boundary}} \\ & + 0.50\mathcal{L}_{\text{semantic-END}} + 0.25\mathcal{L}_{\text{END-margin}}. \end{aligned} \quad (12)$$

For two events sharing a committed prefix, let the student posteriors at protected position i be $\mathbf{p}_i^{(k)}, \mathbf{p}_i^{(k+1)} \in [0, 1]^{V'}$. For the protected-position set $\mathcal{I}$, the commit-consistency loss is

$$\mathcal{L}_{\text{commit}} = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} D_{\text{KL}}(\mathbf{p}_i^{(k)} \parallel \mathbf{p}_i^{(k+1)}). \quad (13)$$

To discourage a premature semantic END, let $l_{\text{end}} \in \mathbb{R}$ be its logit, $l_y \in \mathbb{R}$ the logit of the correct content token, and $m = 2$ the scalar margin. Then

$$\mathcal{L}_{\text{END-margin}} = \max(0, m + l_{\text{end}} - l_y). \quad (14)$$

The loss prefers $l_y \geq l_{\text{end}} + m$ before the content is complete. It does not determine whether the upstream prefix is correct; consequently, it may also prolong an incorrect translation after an irreversible ASR error.

4.6 Proposed RL Stage: Quality-Aware Trajectory-Level GRPO

The previous GRPO route froze the backbone and trained only a binary action head. Such a policy cannot directly repair translation content or semantic completeness. The proposed RL stage starts from Stage B ‘iter_0001132’, or from a later quality-repaired selected checkpoint. The acoustic frontend and BiCodec remain frozen. A top-layer Qwen adapter and action policy are trainable, while content and semantic posteriors are protected with strong reference KL.

For input s , sample G complete trajectories $\{\tau_i\}_{i=1}^G$. Define

$$\begin{aligned} R_i = & w_q Q_i + w_p P_i - w_f \tilde{F}_i - w_a \tilde{A}_i - w_l \tilde{L}_i - w_{\text{pre}} U_i - w_{\text{wait}} W_i \\ & + w_c C_i + w_s S_i + w_b B_i, \end{aligned} \quad (15)$$

where $Q_i, P_i, C_i, S_i, B_i \in [0, 1]$ are normalized bilingual final quality, prefix support, completeness, semantic non-silence, and boundary continuity. $\tilde{F}_i, \tilde{A}_i, \tilde{L}_i \in \mathbb{R}_{\geq 0}$ are normalized First WRITE, ATD, and LAAL; $U_i, W_i \in [0, 1]$ are premature WRITE and unnecessary WAIT. All scalar weights $w. \geq 0$.

The group reward statistics are

$$\mu_R = \frac{1}{G} \sum_{i=1}^G R_i, \quad \sigma_R = \sqrt{\frac{1}{G} \sum_{i=1}^G (R_i - \mu_R)^2}, \quad (16)$$

and the normalized scalar advantage is

$$A_i = \frac{R_i - \mu_R}{\sigma_R + \epsilon}, \quad (17)$$

where $\epsilon > 0$ prevents division by zero. With trainable policy π_θ , frozen reference policy π_{ref} , KL weight $\beta \geq 0$, and replay weight $\lambda_{\text{replay}} \geq 0$, the objective is

$$\mathcal{L}_{\text{GRPO}} = -\frac{1}{G} \sum_{i=1}^G A_i \log \pi_\theta(\tau_i \mid s) + \beta D_{\text{KL}}(\pi_\theta \parallel \pi_{\text{ref}}) + \lambda_{\text{replay}} \mathcal{L}_{\text{replay}}. \quad (18)$$

An RL gain will be accepted only if it outperforms matched continued SFT in both quality retention and latency, and if at least one operating point lies above and to the left of the fixed wait- k quality-latency frontier.

5 Experiments

5.1 Dataset Composition

The principal dataset is public UniST. The training set contains 198 parquet shards and 19,785,924 records; dev contains 7,965 records and test contains 23,369. Each record provides an ID, source and target language, source transcription, target translation, source GLM tokens, source and target BiCodec semantic tokens, and 32 global speaker tokens. Offline sample and packing counts are given in [table 1](#).

Table 1: Offline Phase 1-3 data scale. The Phase 2 ratio denotes new-task examples to Phase 1 replay.

Stage	Unpacked samples	18k packed samples	Task composition
Phase 1	79,143,696	800,632	ASR, S2TT, TTS, and MT proxy
Phase 2 raw	59,357,772	1,968,716	Quality, Performance, and Direct S2ST
Phase 2 + replay	89,036,658		Phase 2:Phase 1 replay = 2:1
Phase 3	39,571,848	1,161,587	Quality and Performance

Streaming Stage A/B uses a fixed 15-shard pilot for controlled iteration. Stage A includes 1,325,243 source records and 16,195 packed training sequences. Stage B expands the same utterances into 25,997,984 training events and 263,391 validation events. Of the training events, 7,666,418 contain a non-empty target WRITE before source EOS. Quality routing produces 385,051 clean, 616,445 noisy-content, and 323,747 quarantine records.

Discussion. The full198 offline data provides broad capability coverage, whereas the fixed 15-shard streaming pilot enables faster causal iteration and matched comparisons. Conclusions about the streaming stages are therefore pilot conclusions, not full-corpus generalization claims.

5.2 Data Construction and a Real Example

Offline preparation maps each parquet record into task-conditioned JSONL and then packs multiple examples into sequences of length 18,000 with reset attention masks. Stage B additionally stores event time, ASR/MT/semantic deltas, alignment confidence, gold/free-running history, and the quality route.

The first record of ‘train-00000.parquet’ is a real English-to-Chinese example:

```
id: NCSSD_R_EN_0000000000
src_lang: eng
tgt_lang: cmn
transcription:
  Hello, I'm looking for a book on mindfulness. Can you help me?
translation (verbatim):
  你好, 我在寻找一本关于正念的书。你能帮我吗?
English gloss:
  Hello, I am looking for a book on mindfulness. Can you help me?
source_glm length: 53
```

```
source_bicodec length: 211
target_bicodec length: 178
bicodec_global length: 32
```

The Chinese reference is stored verbatim in the data; the English gloss above is used only to keep this report fully English. Stage B converts this utterance into 15 streaming events. The first four are summarized in [table 2](#).

Table 2: Stage B trajectory demo constructed from one real UniST record.

Event	Start/ms	End/ms	Source delta/prefix	Target delta	Semantic slice	Confidence
0	0	480	‘Hello’	empty	empty	–
1	480	640	prefix=‘Hello’	‘你好’	[0:28]	0.7227
2	640	800	‘I’m’	empty	empty	–
3	800	960	updated source prefix	‘我在’	[28:40]	0.6443

Discussion. An empty target delta means that the visible source prefix does not yet support a safe translation commit; it is not a missing label. Confidence values near 0.64–0.72 also demonstrate why automatically generated trajectories cannot be treated as perfect ground truth. Quality routing is essential before irreversible supervision is applied.

5.3 Model and Training Configuration

The offline backbone is a 24-layer Qwen2.5-0.5B model with hidden size 896, FFN size 4,864, 14 attention heads, 2 KV heads, sequence length 18,000, and BF16 precision. Megatron-LM runs on one 8-GPU node with tensor parallelism 1, pipeline parallelism 1, micro batch 2, global batch 128, AdamW with $(\beta_1, \beta_2) = (0.9, 0.95)$, and FlashAttention.

Table 3: Training budgets and representative final validation losses. Stage B validation is a post-training fixed-16 free-running gate, not an in-loop held-out loss.

Stage	Updates	Data scope	Representative loss	Initialization / trainable scope
Phase 1	3,300 + 15,465	UniST-198	approximately 5.3098	Full model; low-LR recovery
Phase 2	15,381	UniST-198	approximately 4.2881	Full model from Phase 1
Phase 3	9,075	UniST-198	3.80985	Full model from Phase 2
Stage A	381	15 shards; 3 coverages	numerically stable	Causal frontend, bridge, and shared Qwen
Stage B	2,264	15 shards; 3 coverages	multitask trade-off	Frozen frontend; shared Qwen only

Discussion. The offline runs optimize all model parameters, whereas Stage B deliberately freezes the causal frontend and uses much smaller learning rates for Qwen. This separation improves attribution but does not guarantee ASR retention because the shared decoder posterior still changes.

5.4 Evaluation Metrics

Text quality is measured with corpus Text-BLEU. Speech-BLEU first transcribes generated target speech with a target-language ASR system and then computes BLEU, so it includes translation, synthesis, and evaluation-ASR errors. AutoPCP is an automatic proxy for cross-lingual prosody/expression similarity. UTMOS estimates perceptual naturalness. For source and generated durations $d_s, d_y \in \mathbb{R}_{>0}$ and M evaluated examples, duration consistency is

$$\text{SLC}_\delta = \frac{1}{M} \sum_{i=1}^M \mathbb{I} \left(1 - \delta \leq \frac{d_y^{(i)}}{d_s^{(i)}} \leq 1 + \delta \right), \quad (19)$$

where $\delta \in \{0.2, 0.4\}$ and $\text{SLC}_\delta \in [0, 1]$.

Streaming ASR uses WER or CER. First WRITE and First audio are the consumed source times at the first non-empty target-text commit and first playable non-silent target waveform. If target token j is written at source time $g(j) \in \mathbb{R}_{\geq 0}$, the generated target length is L , the reference length is L^* , and source duration is d_s , the current diagnostic average-token-delay proxy is

$$\text{ATD}_{\text{proxy}} = \frac{1}{L} \sum_{j=1}^L \left[g(j) - \frac{j-1}{\max(L^*, 1)} d_s \right]. \quad (20)$$

The diagnostic utterances do not have strict human word- or audio-level alignments; therefore reported LAAL and ATD values are explicitly proxies and are not directly comparable to standardized SimulEval values. Runtime factor is $\text{RTF} = t_{\text{wall}}/d_s$. Batch-one RTF below 1 means computation is faster than real time, not that the policy emits early.

Discussion. No single metric is sufficient. A lower First audio may indicate a useful early translation, an incomplete output, or an early error. Every latency result must therefore be paired with content, completeness, non-silence, and final-flush measurements.

6 Experimental Results and Discussion

6.1 Offline Phase 1-3: Convergence and Curriculum Gain

Validation loss decreases from approximately 5.31 after Phase 1 recovery to 4.29 after Phase 2 and 3.81 after Phase 3. The Phase 1 recovery removes late high-LR instability. Phase 2 adds direct S2ST with replay, and Phase 3 specializes the task distribution.

Table 4: Aggregate Phase 2-to-Phase 3 changes on UniST dev/test under the same protocol.

Split	Improved cells	Degraded cells	Interpretation
dev	23/24	1/24	Broad improvements in AutoPCP, SLC, and BLEU
test	21/24	3/24	Translation and semantic metrics improve; limited decreases concentrate in UTMOS

Discussion. The improvement is attributable to curriculum and task concentration rather than a new architecture. The 21/24 test result supports the ordering “broad alignment, direct S2ST with replay, final-task specialization.” The remaining UTMOS decreases show that token-level CE does not perfectly represent perceptual naturalness.

6.2 Offline Phase 3: Best UniST Test Results

Table 5: Best Offline Phase 3 checkpoint ‘iter_0009075’ on UniST test.

Mode	Direction	Text-BLEU	Speech-BLEU	AutoPCP	SLC-0.2	SLC-0.4	UTMOS
Performance	ZH→EN	32.4509	19.3770	2.9118	0.6324	0.8782	3.6676
Performance	EN→ZH	40.5404	38.9953	3.2765	0.7184	0.9421	3.3516
Quality	ZH→EN	39.3753	22.7268	2.8850	0.6358	0.8712	3.6680
Quality	EN→ZH	48.1698	46.3063	3.2752	0.7246	0.9349	3.3586

Discussion. Quality mode produces higher Text- and Speech-BLEU in both directions, supporting the value of explicit intermediate text. Performance mode is nevertheless competitive or better on some AutoPCP/SLC cells, showing that a shorter generation path does not uniformly degrade all speech attributes. ZH-to-EN Speech-BLEU remains substantially lower than EN-to-ZH, making English target-speech intelligibility and semantic generalization a primary offline weakness.

6.3 Streaming Stage A: A Working Causal Interface with a Failed Free-Running Gate

Stage A finishes all 381 updates with zero skipped iterations and zero NaNs. It generates pre-final output for 972/972 events, demonstrating that the causal visibility path and event grammar execute correctly. Free-running ASR, however, remains far from the offline anchor.

Table 6: Stage A formal training and free-running ASR results.

Metric	Stage A	Offline anchor	Interpretation
Weighted WER/CER	27.1432%	-	Bilingual aggregate; gate failed
Chinese CER	21.0112%	5.4661%	Large causal-chunk degradation
English WER	35.3399%	5.6072%	More severe English degradation
Causal-full error	14.4546%	-	Complete causal input is easier than chunked free-running
Teacher-forced accuracy	92.0737%	-	Strong prediction under correct history
CTC all-blank cases	15	0 expected	CTC supplies no valid alignment for some samples
Event-stop failures	27	0 expected	Structural stopping failures remain

Discussion. The gap between 92.07% teacher-forced accuracy and 27.14% free-running error indicates exposure-bias and irreversible-prefix accumulation rather than a complete inability to predict ASR tokens. The causal-full result of 14.45% further isolates an additional chunk-boundary/state-propagation penalty. Stage A is therefore a structurally successful but quality-gate-failed causal checkpoint.

6.4 Streaming Stage B: Learned Objectives and Multitask Conflict

Stage B completes three 15-shard coverage epochs and 2,264 updates with no skipped iterations or NaNs. ASR CE, MT CE, and semantic END margin decrease, but replay, KL retention, and boundary objectives do not improve together.

Table 7: Means over the first and last 10% of non-zero Stage B training scalars.

Segment	Metric	Early	Late	Trend
Epoch 1	ASR CE	1.716	0.954	improved
Epoch 1	MT CE	4.699	2.079	improved
Epoch 1	semantic END margin	2.528	0.070	strongly improved
Epoch 1	replay CE	4.018	4.056	flat/slightly worse
Epoch 1	V1 ASR KL	0.255	0.338	retention drift increased
Epoch 2-3	ASR CE	0.818	0.684	improved
Epoch 2-3	MT CE	2.098	1.780	improved
Epoch 2-3	semantic END margin	0.070	0.046	improved
Epoch 2-3	boundary/EOS	0.198	0.534	degraded
Epoch 2-3	content END CE	0.830	1.096	degraded

Discussion. The END-margin objective is clearly learnable, but boundary/EOS and content-END losses move in the opposite direction. V1 ASR KL also rises, showing that freezing the acoustic frontend does not freeze the streaming ASR behavior of the shared decoder. A future supervised revision should adapt retention weights using bilingual free-running gates rather than further increasing END-margin alone.

6.5 Streaming Stage B: Checkpoint Selection and Directional Asymmetry

Table 8: Fixed-16 free-running validation for three Stage B checkpoints. Lower ASR error and higher chrF/coverage are better.

Checkpoint	Chinese ASR error	English ASR error	free ZH→EN chrF	free EN→ZH chrF	semantic coverage	non-silent
iter 1132	0.1174	0.3097	24.078	5.414	0.875	7/8
iter 1207	0.1033	0.3355	21.312	4.827	0.682	6/8
iter 2264	0.0892	0.3032	21.216	4.630	0.797	7/8

Discussion. The final checkpoint has the best ASR retention, but iter 1132 has the best free-running MT and semantic coverage. Continued training therefore improves local ASR while reducing translation quality. All three checkpoints fail the combined structure, coverage, and non-silence gates. Iter 1132 is only the best diagnostic listening candidate among failed checkpoints, not a formally selected model.

6.6 Matched Stage A/B Audio: Latency and Content

The strict causal cascade consumes physical PCM in 160 ms blocks and makes decisions at 160, 320, 640, or 1,280 ms intervals. MT reads only committed ASR, and TTS reads only newly committed target text. Table 9 reports a 4.22-s English-to-Chinese utterance.

Table 9: English-to-Chinese Stage A/B First audio and content. Times are in ms; LAAL is a proxy.

Stage	Chunk	First ASR	First MT	First audio	LAAL proxy	Generated translation
A	160	1120	1600	4220	1378	‘Hello, I’ equivalent
B	160	960	1600	2400	1793	More content, but incomplete
A	320	1280	1920	4220	1698	‘Hello, I’ equivalent
B	320	1280	1920	2880	2033	More content, but incomplete
A	640	1920	3200	4220	2105	Complete and correct
B	640	1920	2560	4220	1885	Complete and correct
A	1280	2560	4220	4220	2332	Complete and correct
B	1280	2560	3840	4220	2181	Complete and correct

The short-chunk First-audio improvements are

$$\Delta F_{160} = 4220 - 2400 = 1820 \text{ ms}, \quad \Delta F_{320} = 4220 - 2880 = 1340 \text{ ms}. \quad (21)$$

These are the clearest latency gains produced by the current streaming innovation, but the corresponding outputs are incomplete. At 640/1280 ms, content is retained while First audio remains at source EOS.

Table 10: Chinese-to-English Stage A/B First audio and generated content for a 9.66-s source.

Stage	Chunk	First ASR	First MT	First audio	Generated translation
A	160	960	1760	2560	‘That actually,’ incomplete
B	160	1120	1440	1760	‘The horse actually waited for us,’ incorrect
A	320	1280	2240	9660	‘That,’ incomplete
B	320	1280	1920	2240	‘The horse ...,’ incorrect
A	640	1920	4480	6400	Complete and semantically reasonable
B	640	1920	2560	3840	‘The horse actually waited for us,’ incorrect
A	1280	2560	6400	7680	Complete and semantically reasonable
B	1280	2560	3840	9660	‘The,’ truncated

Discussion. The lower Chinese-to-English First audio of Stage B is caused by an incorrect irreversible ASR prefix, which is translated as “The horse.” It is not a valid latency improvement. This result demonstrates why simultaneous metrics must be accompanied by semantic correctness and completeness: an early or negative end offset may simply mean that the model speaks incorrectly or stops too soon.

6.7 Earlier Action-Only GRPO: Better Quality but No Latency Gain

The earlier Stage7A comparison evaluates a Stage6 baseline (E0), matched continued SFT (E1), GRPO with group size 4 (E2), and GRPO with group size 8 (E3) on 23,369 test examples. The backbone is frozen and only a WAIT/WRITE action head is trained.

Table 11: Main quality and latency results of the earlier action-only GRPO experiment. This is not RL on the current Stage B model.

Experiment	Text-BLEU ZH→EN	Text-BLEU EN→ZH	First WRITE	ATD	Premature	Unnecessary WAIT
E0 Stage6	26.378	40.560	3986.4	1807.3	0.031	0.158
E1 continued SFT	26.240	40.480	3991.0	1808.4	0.030	0.157
E2 GRPO G4	27.587	40.224	4055.4	1839.0	0.027	0.164
E3 GRPO G8	28.109	40.355	4032.8	1827.5	0.029	0.162

Relative to matched E1, E3 improves ZH-to-EN Text-BLEU by $28.109 - 26.240 = 1.869$ but increases First WRITE by 41.85 ms and ATD by 19.14 ms. GRPO reduces a small amount of premature WRITE while increasing unnecessary WAIT.

Discussion. This experiment is a direct counterexample to the claim that adding RL necessarily reduces latency. Without explicit First WRITE/ATD, bilingual quality, completeness, and audio-validity terms, the group-relative objective prefers safer but slower trajectories. This failure motivates the multi-component reward in [equation \(15\)](#).

6.8 Reward-v2 Evidence and the Acceptance Criteria for Future RL

Reward-v2 experiments R0–R3 add coverage, explicit latency, and bilingual/adaptive-KL terms on the older Stage6 action-head route. They are preliminary evidence and must not be attributed to Stage B.

Table 12: Reward-v2 dev results. R1 is the matched rebalanced-plus-coverage control.

Experiment	ZH→EN Text-BLEU	EN→ZH Text-BLEU	First WRITE	ATD	Premature	Unnecessary WAIT
R0 E3-v1+bias	27.592	37.724	4205.6	1890.0	0.032	0.150
R1 coverage	28.952	37.610	4234.1	1907.9	0.031	0.149
R2 explicit latency	28.983	37.458	4111.2	1861.4	0.037	0.141
R3 bilingual+adaptive KL	29.277	37.694	4157.5	1883.2	0.035	0.144

Relative to R1, R2 reduces First WRITE and ATD by 122.9/46.5 ms while reducing EN-to-ZH Text-BLEU by 0.152. R3 reduces First WRITE and ATD by 76.6/24.7 ms and improves the two Text-BLEU directions by 0.325/0.084.

Discussion. These trends support bilingual quality terms and adaptive KL, but they remain single-seed dev evidence. A future Stage B RL study must include matched continued SFT, the fixed wait- k frontier, at least three seeds, 10,000 paired-bootstrap samples, batch-one latency, both Text- and Speech-BLEU directions, UTMOS/AutoPCP, premature/unnecessary WAIT, final flush, completeness, and non-silence.

7 Ablation Interpretation and Error Analysis

7.1 Value and Limits of Replay

The Phase 2 replay plus Phase 3 specialization pipeline improves 21 of 24 principal test cells, supporting replay as an effective offline curriculum mechanism. In Stage B, however, replay CE is nearly flat and V1 ASR KL increases. Fixed weights of 0.50 for replay and 0.30 for ASR KL are therefore insufficient to prevent posterior drift in the shared Qwen. Future retention weights should respond to bilingual free-running gates instead of only to training CE.

7.2 Scope of the Chunk Curriculum

Multi-chunk training allows one checkpoint to support 160–1,280 ms operating points rather than requiring a separate model for each delay. Short chunks nevertheless contain less semantic evidence and are more likely to cause permanent prefix errors. The combination of earlier incomplete English-to-Chinese speech and early incorrect Chinese-to-English speech shows that chunk curriculum must be paired with prefix-support supervision.

7.3 The Double-Edged Effect of END Margin

Semantic END margin decreases from 2.528 to 0.070, proving that the objective is learnable, while boundary/EOS and content-END CE degrade later. END margin answers whether generation should continue, not whether the current content is correct. When ASR is wrong, a stronger margin can make the wrong semantic sequence longer and more confident. It should be gated by translation coverage, source-prefix support, and teacher posterior reliability.

7.4 Speaker Consistency and Audio Continuity

The Stage B speaker-continuity weight is zero because no reliable cross-fragment speaker-embedding sidecar is available. Offline Phase 3 conditions synthesis on 32 global BiCodec tokens from the source speaker. Keeping these tokens fixed helps preserve global identity, but fragmented semantic generation, short codec context, and incorrect language content can still change perceived timbre. Future work should add genuine cross-fragment speaker-embedding consistency and boundary cross-fade/holdback evaluation; AutoPCP alone is not proof of speaker identity.

8 Limitations and Threats to Validity

- The offline results are a public UniST-198 reproduction with a 0.5B backbone, not an exact reproduction of the original UniSS 1.5B system and its complete corpus. The numbers must not be presented as a direct paper ranking.
- Stage A/B are trained on a fixed 15-shard pilot. Stage B selection uses a fixed-16 free-running gate, and the matched audio latency analysis contains only two representative utterances. These results do not establish population-level significance.
- Stage A teacher KL has a zero effective denominator, while cache/full consistency is an external gate. Neither can be claimed as an experimentally validated loss contribution.
- Stage B has no conventional in-loop held-out validation loss. Its validation values are post-training fixed-16 free-running gates.
- LAAL and ATD are proxy values derived from causal commit times without strict human source-target alignment. They cannot be directly ranked against standardized values reported by other papers.
- E0–E3 and R0–R3 use a frozen Stage6 action head and are not full content/semantic RL on Stage B. They are evidence for reward design only.
- Stage B’s short-chunk English-to-Chinese First-audio gains are accompanied by incomplete content, while earlier Chinese-to-English output is incorrect. We claim a local first-packet improvement, not a bidirectional overall S2ST success.

9 Conclusion and Next Steps

This report provides a unified account of Offline Phase 1-3 and Streaming Stage A/B. The three-phase offline curriculum trains stably on the full public data and produces the current best checkpoint, Phase 3 ‘iter_0009075’, which improves 21 of 24 principal UniST test cells over Phase 2. Stage A establishes a causal PCM-to-acoustic-embedding-to-append-only-ASR path but does not pass the free-running ASR gate. Stage B jointly trains incremental ASR, MT, semantics, and replay in the shared Qwen. It reduces English-to-Chinese First audio by 1.82/1.34 s at 160/320 ms chunks, but exposes severe directional asymmetry and irreversible-error propagation.

The next step should not be an immediate full-data expansion or a repetition of action-only GRPO. The recommended sequence is: (1) repair Chinese-to-English irreversible ASR/MT errors using stronger free-running retention and prefix-support objectives; (2) add a strict validation split, larger bidirectional causal evaluation, and non-proxy alignment; (3) apply the quality-aware GRPO objective in [equations \(15\) and \(18\)](#) to a genuinely selected Stage B checkpoint; and (4) expand to full198 only after bilingual quality, complete non-silent speech, and latency improve together. The central methodological contribution is not the concealment of failed gates, but the separation of offline quality, causal executability, and true Pareto improvement within one auditable model and data pipeline.

A Reproducibility Assets

Asset	Repository-relative path
Offline overview	docs/uniss_training_reproduction/uniss_project_offline_phase1_3_and_streaming_stage_a_b_colleague_overview.md
Phase 2/3 evaluation	docs/uniss_training_reproduction/uniss_full198_phase2_phase3_detailed_evaluation_report.md
Stage A result	reports/uniss_phase3_v4_quality_first_true_streaming_pilot15_v1/stage_a_formal/stage_a_formal_20260816T224100Z/STAGE_A_RESULT_REPORT.md
Stage B implementation audit	docs/uniss_training_reproduction/uniss_phase3_v4_e2e_simuls2st_native_megatron_implementation_report.md
Stage A/B audio evaluation	eval_outputs/uniss_phase3_v4_e2e_simuls2st_pilot15_v1/stage_b_endmargin_best_iter1132_stage_a_protocol_20260825T114500Z/STAGE_B_VS_STAGE_A_STREAMING_EVALUATION_REPORT.md
Earlier GRPO test	eval_outputs/simul_uniss_stage7a_15shard_v1/full_test_e2e_v1/stage7a_four_way_full_test_report.md
Reward-v2 dev	eval_outputs/simul_uniss_stage7a_reward_v2_15shard_v1/full_dev_e2e_v1/reward_v2_four_way_full_dev_report.md
Phase 3 checkpoint	checkpoints/uniss_qwen0p5b_phase3_unist198_after_phase2_v4/iter_0009075

Asset	Repository-relative path
Stage A checkpoint	checkpoints/uniss_phase3_v4_quality_first_true_streaming_pilot15_v1/stage_a_formal/stage_a_formal8_20260816T224100Z/iter_0000381

B Status Boundary of the Claims

Table 14: Completion status and the strongest defensible claim for each component.

Component	Status	Defensible claim
Offline Phase 1-3	✓	Full198 training complete; Phase 3 is the current best offline checkpoint
Streaming Stage A	×	Structure and numerics work; free-running ASR gate fails
Streaming Stage B	×	Three coverages complete; local EN-to-ZH latency gain, combined bilingual gate fails
Earlier action-only GRPO	✓	Experiment complete; does not improve the matched-SFT quality-latency Pareto point
Reward-v2 action head	✓	Positive dev trend; not RL on the current Stage B model
Quality-aware Stage B GRPO	Proposed	Not yet executed; no gain may be claimed

References

- [1] Sitong Cheng, Weizhen Bian, Xinsheng Wang, Ruibin Yuan, Jianyi Chen, Shunshun Yin, Yike Guo, Wei Xue, et al. UniSS: Unified expressive speech-to-speech translation with your voice. *arXiv preprint arXiv:2509.21144*, 2025.
- [2] Mingbo Ma, Liang Huang, Hao Xiong, Renjie Zheng, Kaibo Liu, Baigong Zheng, Chuanqiang Zhang, Zhongjun He, Hairong Liu, and Xing Li. STACL: Simultaneous translation with implicit anticipation and controllable latency using prefix-to-prefix framework. *arXiv preprint arXiv:1810.08398*, 2019.
- [3] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. *arXiv preprint arXiv:2212.04356*, 2022.
- [4] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, et al. DeepSeekMath: Pushing

the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.

- [5] Mohammad Shoeybi, Mostofa Patwary, Raul Puri, Patrick LeGresley, Jared Casper, and Bryan Catanzaro. Megatron-LM: Training multi-billion parameter language models using model parallelism. *arXiv preprint arXiv:1909.08053*, 2019.
- [6] Xinsheng Wang et al. Spark-TTS: An efficient llm-based text-to-speech model with single-stream decoupled speech tokens. *arXiv preprint*, 2025.
- [7] An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.
- [8] Shaolei Zhang, Qingkai Fang, Shoutao Guo, Zhengrui Ma, Min Zhang, and Yang Feng. StreamSpeech: Simultaneous speech-to-speech translation with multi-task learning. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024.